

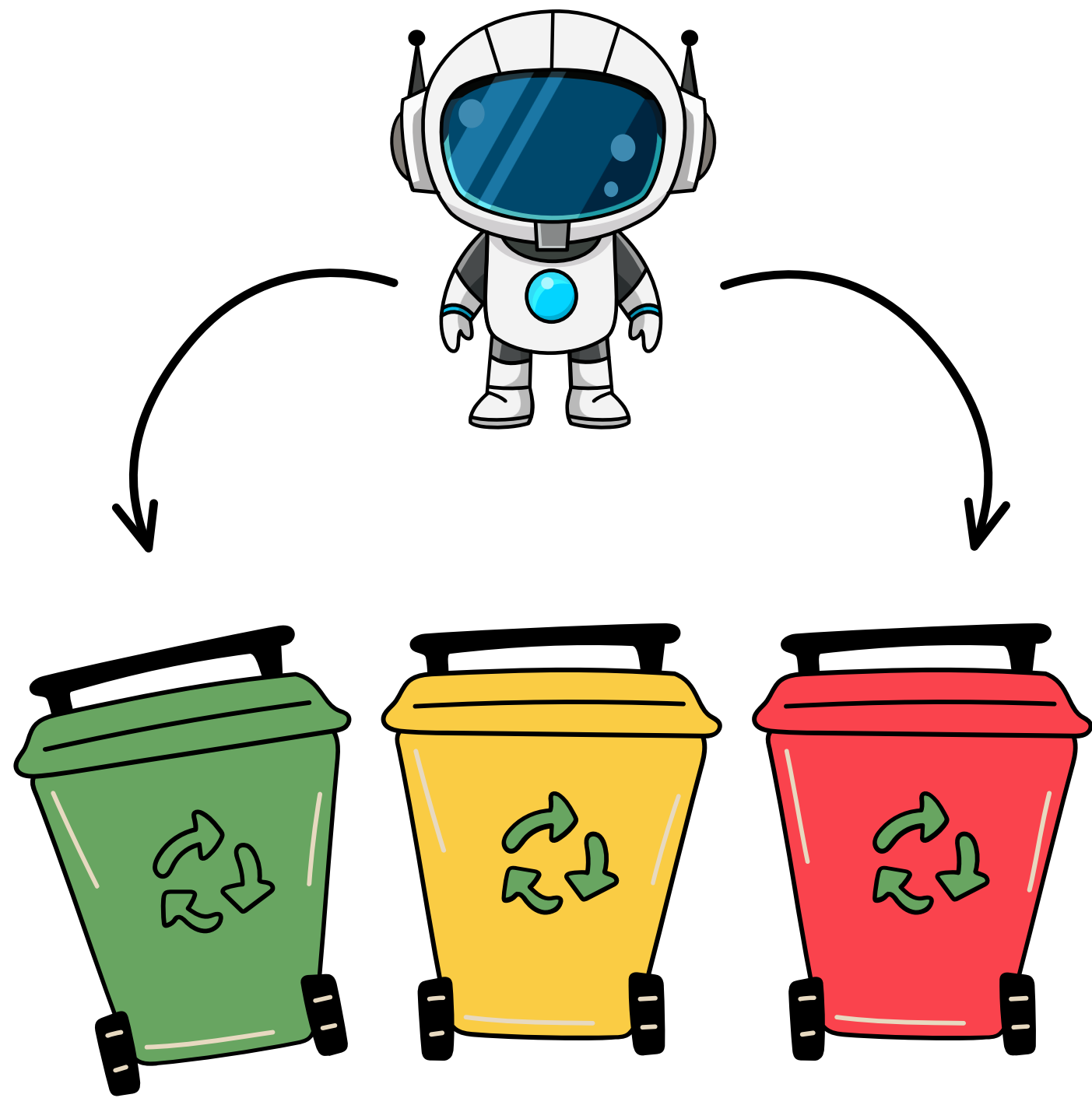
ReLoop AI

AI-Powered Circular Economy Platform

Transforming Waste into Value Through a
Circular Economy

Presented by : Divya J & Deepika S





Introduction

What is ReLoop AI?

- ReLoop AI is a digital platform that redefines how waste is perceived and managed.
- Instead of treating waste as a disposal problem, ReLoop AI treats waste as a resource with measurable value.
- The platform enables individuals, communities, recyclers, NGOs, and industries to participate in a shared circular economy ecosystem where materials are reused, recycled, and reintegrated into the supply chain.

Mission Statement

Our mission is to build a future where waste no longer exists as an end product.

“Every discarded item deserves a second life.”

ReLoop AI aims to empower people with knowledge, tools, and digital infrastructure to make sustainable waste decisions and contribute to a circular economy.





Problem Statement

The Global Waste Challenge

Despite increasing awareness about sustainability, millions of tons of recyclable and reusable materials still end up in landfills every year.

This happens because:

- People are unaware of the value of their waste
- Recycling information is scattered and inconsistent
- Industries struggle to source recycled materials efficiently
- Small quantities of waste fail to reach large recyclers
- There is no single platform connecting waste generators and resource consumers

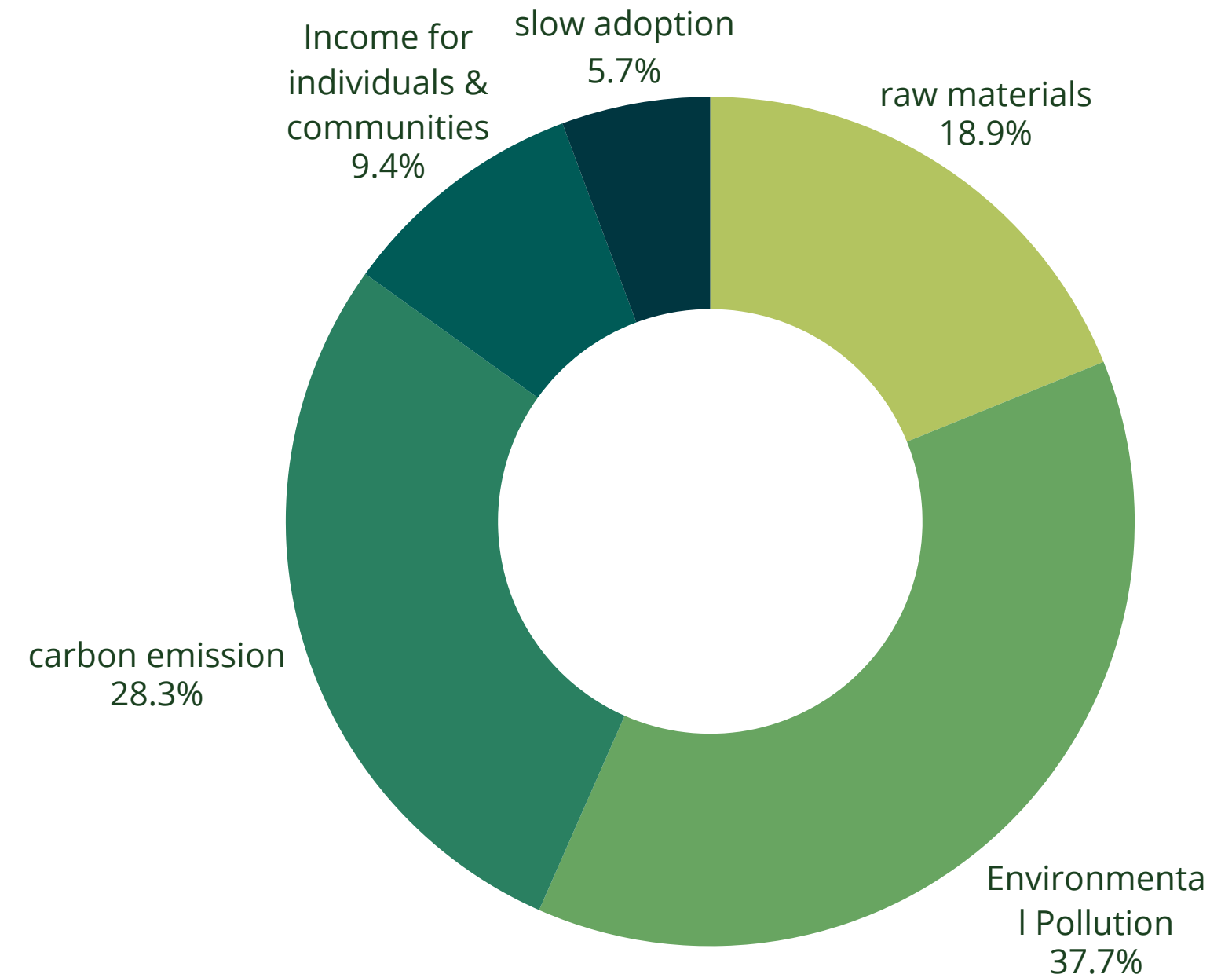


Consequences of the Problem

Why the Current System Fails

The lack of a connected circular system results in:

- Loss of valuable raw materials
- Increased environmental pollution
- Higher carbon emissions
- Missed income opportunities for individuals and communities
- Slow adoption of circular economy practices
- High percentage of recyclable waste sent to landfills

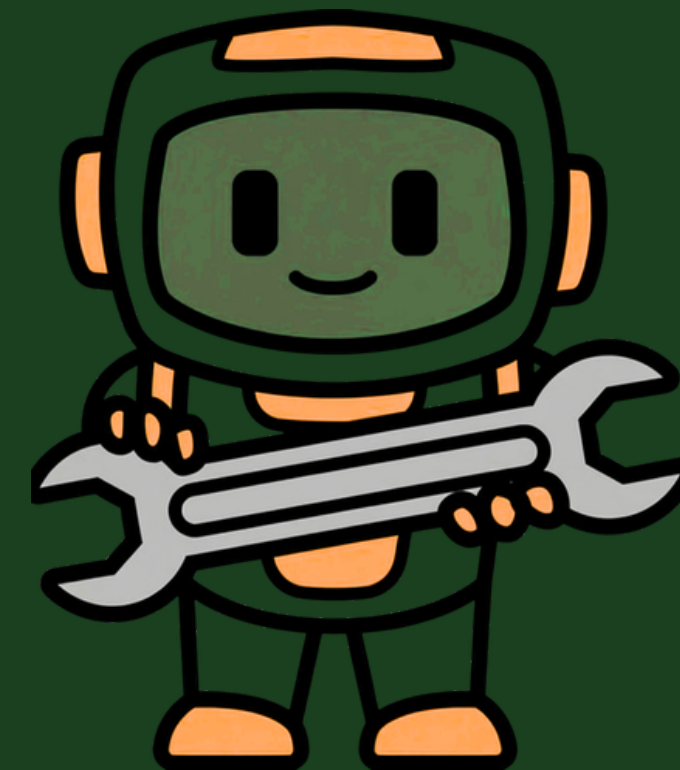
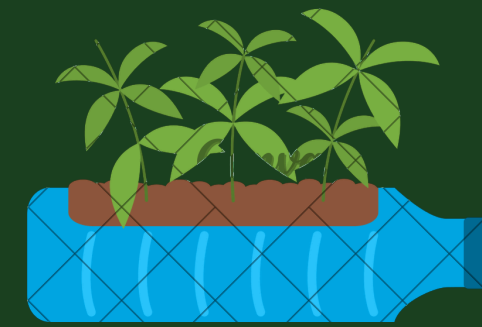


This chart shows how inefficient waste handling leads to environmental damage and economic loss when recyclable materials are not recovered.

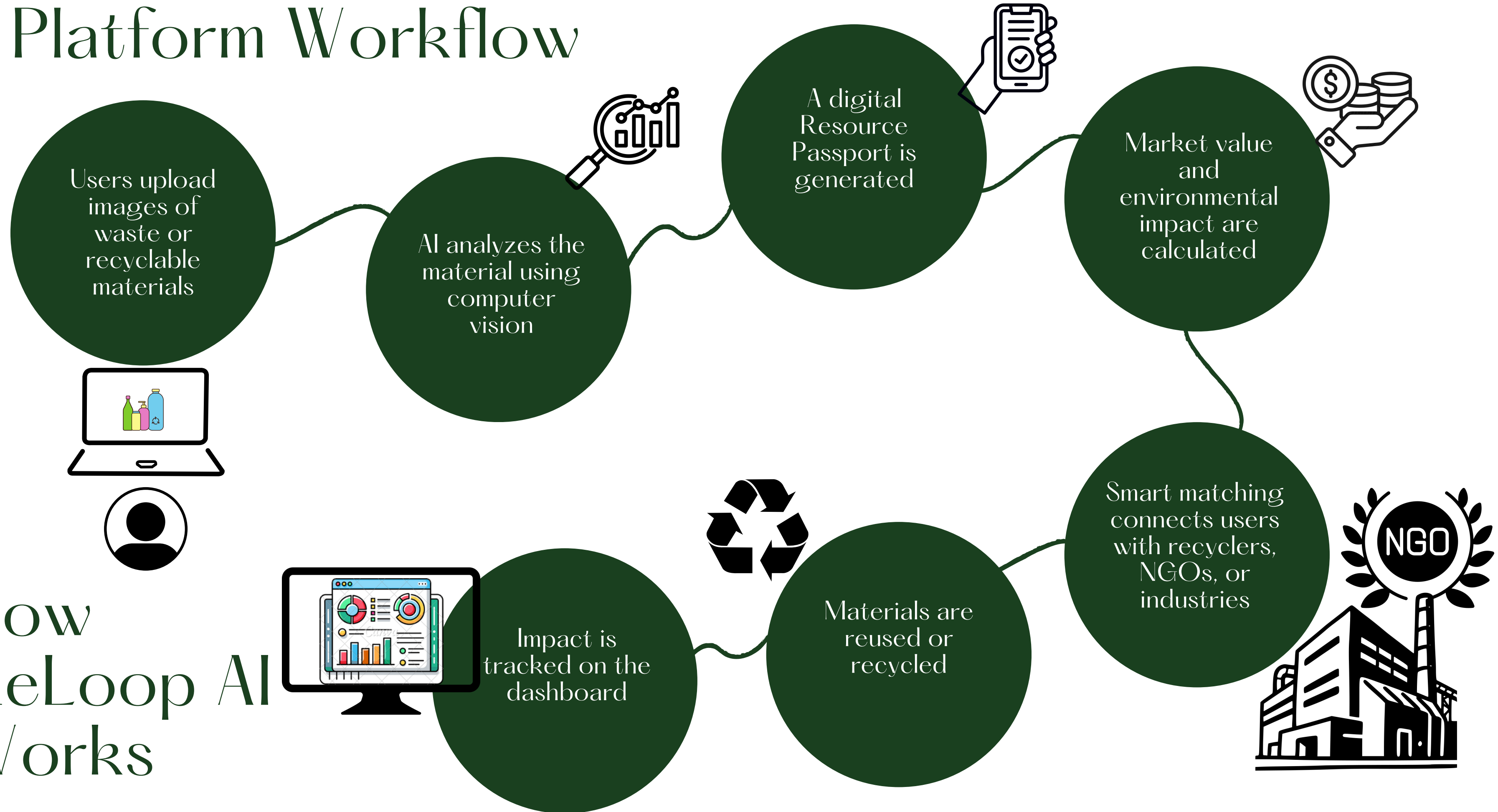
Proposed Solution

How ReLoop AI Solves This

- ReLoop AI introduces a unified digital ecosystem where waste materials are analyzed, valued, and redirected to the most suitable reuse or recycling pathway.
- By combining AI, marketplaces, and smart matching, ReLoop AI ensures that waste is not discarded blindly but handled intelligently.



Platform Workflow



How
ReLoop AI
Works

AI Resource Scanner

AI-Powered Resource Scanner

The AI Resource Scanner is the core intelligence layer of ReLoop AI. Users can upload images of waste materials using a mobile phone or web application. The system uses computer vision and machine learning models to analyze the uploaded image.

The scanner identifies:

- Material type (plastic, metal, paper, etc.)
- Object category
- Approximate quantity
- Physical condition
- Recyclability and reusability potential

This feature removes uncertainty for users and enables accurate, data-driven waste classification without requiring expert knowledge.



Resource Passport (Data & Value Layer)

Digital Resource Passport

Once a material is scanned, ReLoop AI generates a Digital Resource Passport that acts as a unique identity card for the material.

The Resource Passport contains:

- Material composition details
- Circularity score indicating sustainability potential
- Reuse and recycling possibilities
- Estimated market value
- Environmental impact metrics such as CO₂ savings
- Recommended next action (reuse, donate, recycle, or sell)

This passport ensures transparency, traceability, and informed decision-making across the circular economy network.



Circular Marketplace & Smart Matching

ReLoop AI provides a digital circular marketplace where waste materials are treated as tradable resources.

Users can:

- Sell recyclable materials
- Donate reusable items
- Exchange materials within communities
- Connect with verified buyers and recyclers

The Smart Matching Engine automatically connects users with the most suitable partners based on:

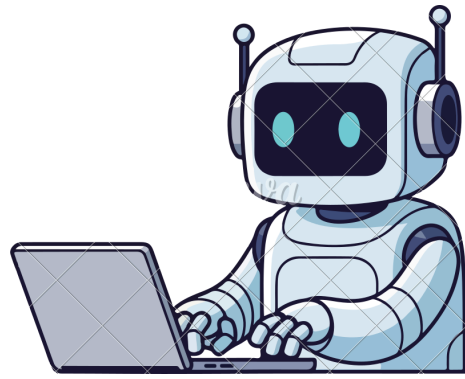
- Material type
- Quantity available
- Geographic location
- Current market demand
- Pickup and logistics availability

This integration ensures faster transactions, reduced waste leakage, and efficient material circulation.



Together we can recycle much more

Conclusion



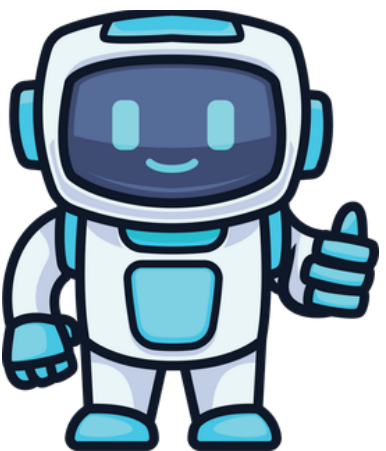
ReLoop AI provides a practical solution to the problem of ineffective waste management by using artificial intelligence to convert waste into valuable resources.

Instead of treating waste as something to discard, the platform helps reuse and recycle materials through a circular economy approach.

By using AI-based material identification, digital resource passports, smart matching, and a circular marketplace, ReLoop AI connects individuals, communities, recyclers, NGOs, and industries on a single platform.

This helps reduce landfill waste and improves material recovery.

Overall, ReLoop AI shows how technology can support sustainability by reducing pollution, creating economic value, and encouraging responsible waste management, where every discarded item gets a second life.





Thank You

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